

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	(a)			(enthalpy change) when 1 mol (of substance) is burned (1) in excess oxygen (1)	2			2		
	(b)	(i)	I	$n(\text{propanol}) = \frac{0.14}{60} = 0.00233$		1		1	1	1
			II	$\Delta T = 12.5$ (1) energy = $mc\Delta T = 100 \times 4.18 \times 12.5 = 5225 \text{ J}$ (1) $\Delta H = -\frac{5.225}{0.00233} = -2242 \text{ kJ mol}^{-1}$ (1)		3		3	3	2
		(ii)		credit possible for agree / disagree award (1) for any of following agree – because the temperature rise will be greater (making the percentage error lower) agree – because greater mass of fuel burned (making the percentage error lower) disagree – because more heat will be lost to the surroundings (making ΔT value less accurate)			1	1		1

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	(c)			bonds broken $3(\text{C—C}) + 9(\text{C—H}) + (\text{C—O}) + (\text{O—H}) + 6(\text{O=O})$ $1044 + 3708 + (\text{C—O}) + 463 + 2976$ $8191 + (\text{C—O})$ (1) bonds formed $8(\text{C=O}) + 10(\text{O—H})$ $6440 + 4630$ 11070 (1) enthalpy change $8191 + (\text{C—O}) - 11070 = -2676$ (1) $(\text{C—O}) = -2676 - 8191 + 11070$ $(\text{C—O}) = 203$ (1) ecf possible throughout						
				Question 12 total	2	8	1	11	6	4